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Partial Measurement of Quantum Circuit in Qiskit

Asked 3 years, 6 months ago Modified 3 years, 6 months ago Viewed 1k times



I want to do partial measurement of quantum circuit containing two qubits in entangled state. I want to first measure one qubit and record the result. Thereafter, record the measurement of second qubit and compare the two.

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Mathematically, the two should come out to be same.



How to do the same on Qiskit using Python?



qiskit

programming

measurement



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edited Oct 23, 2020 at 18:57

asked Oct 23, 2020 at 7:18



luciano

5,763 1 13 34



prn

21 2



By the way, the state $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ is an entangled state. And if you measure the first qubit and get a $|0\rangle$ then the state has collapsed onto the state $|01\rangle$ so the second qubit measurement now will give you a $|1\rangle$. – [KAJ226](#) Oct 23, 2020 at 15:38



With reference to comment by KAJ266, the results of measurement are the same in case of Bell state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ or GHZ state $\frac{1}{\sqrt{2}}(|0\dots 0\rangle + |1\dots 1\rangle)$ but not generally. – [Martin Vesely](#) Oct 24, 2020 at 6:28

1 Answer

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Consider the maximal entangled state

3

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$



If I make a measurement on the first qubit and a zero is returned then this implies my state has collapsed into the eigenvector $|00\rangle$ and so the second qubit measurement will definitely returned a $|0\rangle$ as that the only possibility.



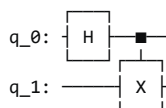
However, if you consider the another entangled state

$$|\phi\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

then measuring the first qubit and receiving a $|0\rangle$ means the quantum state has collapsed onto the eigenstate $|01\rangle$, and so measuring the second qubit will now definitely returned a $|1\rangle$ as that the only possibility.

If you want to extract the quantum state after making measurement on the first qubit before measuring the second qubit in Qiskit, you can use the function `get_statevector()` in the Aer's `statevector_simulator`.

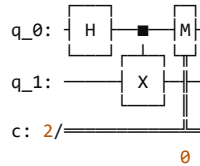
So for example, suppose I want to do this on the state $|\psi\rangle$ defined above, which can be implemented on the quantum circuit as



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```
backend = Aer.get_backend("statevector_simulator")
qc = QuantumCircuit(2, 2)
qc.h(0)
qc.cx(0,1)
qc.measure([0], [0])
print(qc)
result = execute(qc, backend=backend, shots=1).result()
print('State after first measurement:', result.get_statevector() )
```



State after first measurement: $[1.+0.j \ 0.+0.j \ 0.+0.j \ 0.+0.j]$

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edited Oct 23, 2020 at 18:57



luciano

5,763 1 13 34

answered Oct 23, 2020 at 16:08



KAJ226

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